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Why did Ancient States strike Coins?

CHRISTOPHER J. HOWGEGO

IN an influential article published in 1970, Crawford argued that the only reason for the striking of coinage in the ancient world was to make state payments.[1] This view was expressed uncompromisingly:

> It is said on the evidence of Statius that the Emperors took thought for the supply of coinage to the whole Empire. It is also said that coined money had in the ancient world an economic reason for existence. Neither statement is true. Coinage was probably invented in order that a large number of state payments might be made in a convenient form and there is no reason to suppose that it was ever issued by Rome for any other purpose than to enable the state to make payments, that is, for financial reasons...
>
> And we have seen that in the cities of the Roman Empire, in striking contrast to the cities of the pre-Greek East, it [coinage] acquired an important rôle as a means of exchange. But this monetary, economic function, like the other monetary functions of coinage, was an accidental consequence of the existence of coinage, not the reason for it.

Elsewhere Crawford claims that this article contains 'the general principles according to which Roman monetary history should be interpreted'.[2] The article was important in giving impetus and direction to the shift in numismatic studies away from political aspects and towards financial ones and its impact is apparent in the surprise now evinced when patterns of coin production do not appear to match patterns of expenditure. Yet it ought to raise alarm bells that Strabo, when describing the customs of the Persians in a passage which apparently draws on a source of the third century BC, found

I am very grateful to John Griffith, Barbara Levick, Fergus Millar, Keith Rutter and David Walker for comments on an earlier version of this paper and to Cathy King for discussions on the monetary history of the fourth century AD.

Abbreviations: Austin, *The Hellenistic World*: M. M. Austin, *The Hellenistic World from Alexander to the Roman Conquest* (Cambridge, 1981); Austin and Vidal-Naquet, *ESH*: M. M. Austin and P. Vidal-Naquet, *Economic and Social History of Ancient Greece: an Introduction* (1977); Crawford, *CMRR*: M. H. Crawford, *Coinage and Money under the Roman Republic. Italy and the Mediterranean Economy* (1985); Howgego, *GIC*: C. J. Howgego, *Greek Imperial Countermarks. Studies in the Provincial Coinage of the Roman Empire* (1985); Millar, *ERW*: F. Millar, *The Emperor in the Roman World* (1977); Walker, *Metrology*: D. R. Walker, *The Metrology of the Roman Silver Coinage*, parts 1–3, *BAR Supplementary Series* 5; 22; 40 (Oxford, 1976–8).

[1] M. Crawford, 'Money and exchange in the Roman world', *JRS* 60 (1970), pp. 40–8; restated in id., *La Moneta in Graecia e a Roma* (Rome, 1982), pp. 120–2.

[2] id., 'Finance, coinage and money from the Severans to Constantine', *ANRW* II.2, pp. 560–93, at p. 561, n. 4.

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it a cause for comment that the Persians coined only so much money as was commensurate with their expenditures.[3] If this was the normal practice of Greek cities and of Rome why is it singled out as a Persian custom? The sole evidence cited by Crawford to support his view was the supposedly close correlation between the output of silver coinage and military expenditure during the Roman Republic. Although this close correlation was argued to exist only for a restricted period (*c.* 157 BC to *c.* 80 BC, with the exception of the 90s) and largely ignored non-military expenditure, it was nevertheless a powerful argument.[4] It can, however, no longer be accepted. Not only has the correlation on the basis of Crawford's figures now been shown to be weak but his figures for military expenditure are speculative and his estimates of coinage output have been undermined.[5] His view thus stands as influential but without any supporting evidence.

The issues raised by Crawford are so central to our understanding of the historical importance of ancient coinage that his thesis has naturally not passed without comment. Three dissenting voices may be noted in particular.[6] Lo Cascio advanced a 'modernizing' reconstruction of Roman monetary policy which canvassed possibilities such as deliberate 'control of the money supply, in order to have a positive influence on production and exchange', coinage policies intended to manipulate the relative prices of different metals, or the use of revaluations to preserve liquidity. Such theories run beyond the ancient evidence and are a matter of taste. Howgego argued that the model of state expenditure as an explanation for coinage was unsatisfactory in the case of Roman provincial coinage. Finally, Burnett has pointed out that coinage connected with the needs of maintaining a closed currency system or with a monetary reform falls outside Crawford's explanation. There has, however, been no systematic discussion or rebuttal of Crawford's view for the ancient world as a whole and much relevant ancient evidence has not been brought to bear on the topic.

It is argued here that the reasons for coining were more complex than Crawford allowed. Much of the evidence cited will be literary (including inscriptions, papyri and law codes) because it is in these sources that motivations are sometimes explicit. Even when literary sources describe events which are unhistorical they may provide evidence for what seemed

[3] ap. Strabo xv, 3, 21 (*F.Gr.H.* 128 F3a). The Polycritus cited by Strabo is probably a mistake for Polycleitus, who wrote history and geography in the early third century BC. cf. Herodotus iii, 96.

[4] Crawford, *RRC*, pp. 696–707.

[5] B. Frier, *Phoenix* 30 (1976), pp. 375–82; *QT* 10 (1981), pp. 285–95; H. B. Mattingly, *NC* 137 (1977), pp. 199–215; K. Hopkins, *JRS* 70 (1980), p. 111; T. V. Buttrey, *CP* 84 (1989), pp. 73–4; and especially Buttrey, forthcoming (based on the paper read to the Royal Numismatic Society on 18 October 1988).

[6] E. Lo Cascio, 'State and coinage in the late Republic and early Empire', *JRS* 71 (1981), pp. 76–86; Howgego, *GIC*, p. 89; A. Burnett, *Coinage in the Roman World* (1987), pp. 88–92, 'Reasons for minting'.

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plausible to their author and hence reveal aspects of the world in which he lived. The coinage itself rarely records the reason for its issue and too great a reliance on the conclusions of modern scholars about why certain coinages were struck runs the risk of foisting modern notions upon antiquity.

Nothing will be said about the reasons for the development of the earliest electrum coinages as it is not possible to progress beyond defining limits for speculation within our understanding of their political and social context.[7] It is, however, worth recalling two important observations by Kraay about Greek silver coinages in the sixth and fifth centuries BC which are perhaps in need of some modification.[8] First, limited circulation patterns were taken to show that external trade could not have been a major factor in the decision to coin except in the case of cities with ready access to a supply of silver (the coins of Athens and the Thraco-Macedonian area were exported in quantity). Since this point was made new hoards have somewhat changed our view of the extent to which some coinages travelled. We now know, for example, that significant quantities of Corinthian and Aeginetan coin reached Sicily by the end of the sixth century and Egypt by the third quarter of the fifth.[9] Furthermore, the apparent failure of some coins to travel widely does not prove that they were not used for trade – the coins might have been melted down on arrival or traded back to their city of origin – and anyway a regional pattern of circulation (beyond the city of origin) is fairly typical and needs explaining somehow. Kraay's second observation was that the minimal supply of low denominations in all but a few states made it implausible to view the needs of retail trade as a prime determinant. This argument has been somewhat weakened by the realisation that the quantity of small change has been significantly underestimated because of the avoidance by collectors of smaller denominations, the failure of scholars to include them in their studies, the infrequency with which they were hoarded, and the greater difficulty of recovering them from the earth.[10] We should, therefore, be more tentative than before in accepting Kraay's observations. In any case, although the failure of many coinages to be exported in quantity and the apparent shortage of small denominations would be consistent with Crawford's view that coinage was struck to meet expenditure, they by no means exclude other types of motivation. It is also worth stressing that, whatever the reasons why coinage began to be struck and spread across the

[7] M. Price, 'Thoughts on the beginnings of coinage' in *Studies in Numismatic Method Presented to Philip Grierson*, edited by C. Brooke *et al.* (Cambridge, 1983), pp. 1–10; C. M. Kraay, *Archaic and Classical Greek Coins* (1976), pp. 317 ff.

[8] C. M. Kraay, 'Hoards, small change and the origin of coinage', *JHS* 84 (1964), pp. 76–91.

[9] C. Arnold-Biucchi, L. Beer-Tobey and N. M. Waggoner, 'A Greek silver hoard from Selinus', *ANSMN* 33 (1988), pp. 1–35; M. Price and N. Waggoner, *Archaic Greek Coinage. The Asyut Hoard* (1975).

[10] D. Bérend, 'Réflexions sur les fractions du monnayage grec' in *Festschrift für Leo Mildenberg* (edited by A. Houghton *et al.*), (Wetteren, 1984), pp. 7–30.

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Greek world, it would not be surprising if other uses and advantages soon became clear.

From the better known periods of Greek and Roman history a framework of possibilities can be built up and illustrated. Although not all possibilities are applicable to all periods, a long chronological span is useful both because evidence for particular periods is very patchy and because Crawford claimed his view held good for the ancient world as a whole.

BULLION FOR COINAGE

An unfortunate side-effect of concentration on the relationship of coinage with expenditure has been to divert attention away from the influence of the supply of bullion on coinage output. Of course there was no automatic requirement to coin available bullion and one could suppose that the motivation to coin came from the ends rather than the means. However, not only was bullion a *sine qua non* for coinage which might significantly affect patterns of output, but the availability of metal posed questions about how it was to be used. So when the Athenians amassed a large sum from a rich strike of silver at Laurium it was proposed to share it out among themselves at ten drachms a man until Themistocles persuaded them to use it to construct two hundred warships.[11] In either case it might have been coined but it was the bonanza of silver rather than the need to make a specific expenditure that set events in motion. The productivity of the mine is reflected in Athenian coin of Asyut Group IV, 'probably representing one of the most intensive periods of minting in the history of Greek coinage'.[12]

Bullion direct from the mines was an important source of metal for coinage. The substantial issues of silver in large denominations by tribes in Macedonia and Thrace in the late sixth and first half of the fifth centuries BC, much of which found its way to the Levant and Egypt, can only be understood by the need to exploit local mines for trade.[13] The dating of substantial issues down to the middle of the fifth century no longer permits explanations concerned solely with tribute for Persia.[14] In the same way gold mines in the territory of Krenides lay behind the beginning of the massive gold coinages in the name of Philip of Macedon.[15] It is hard to dissociate the resumption of a regular silver coinage at Rome in 157/156 BC after a break

[11] Herodotus vii, 144, 1–2.

[12] Price and Waggoner, *Asyut*, p. 63.

[13] Kraay, *JHS* 84 (1964), pp. 82–3; N. Gale, W. Gentner and G. Wagner, 'Mineralogical and geographical silver sources of archaic Greek coinage', and M. Price, 'The uses of metal analysis in the study of archaic Greek coinage: some comments', in edd. D. M. Metcalf and W. A. Oddy, *Metallurgy in Numismatics* (1980), vol. 1, pp. 3–54; Herodotus v, 17; 23; vi, 46–7; vii, 112.

[14] Price and Waggoner, *Asyut*, pp. 29; 42; 120; J. H. Kagan, 'The Decadrachm hoard: chronology and consequences' in *Coinage and Administration in the Athenian and Persian Empires*, *BAR International Series 343* (Oxford, 1987), edited by I. Carradice, pp. 21–8, at pp. 24–5; M. J. Price, 'The coinages of the northern Aegean', ibid., pp. 43–51.

[15] Diodorus xvi, 8, 6–7.

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since *c.* 170 BC from the re-opening (or opening) of mines in Macedonia in 158 BC.[16] By contrast, the disruption of the mining settlement at Rio Tinto *c.* AD160–70 may have been a significant reason for the apparent decline in production of denarii under Commodus and for the increasing rapidity of debasements of the silver coinage from the beginning of the reign of Marcus Aurelius.[17]

Bullion did not have to come direct from the mines: the use of booty, indemnities, gifts and stored-up wealth are adequately attested. Booty from the Persian Empire, available in quantity from 333 BC, lay behind the massive coinages of Alexander.[18] Diodorus tells us that after the Phocians seized the sanctuary at Delphi in the Third Sacred War they struck gold and silver coins from the offerings to the gods in order to pay mercenaries in 353/352 BC.[19] A decade earlier the Arcadians had made similar use of the temple treasures of Olympia.[20] The indemnity of 2,000 talents of silver imposed on the Carthaginians in 480 BC after their defeat at Himera led to massive issues of silver coins at Syracuse, and Diodorus implies that the Syracusans also used the crown of 100 gold talents from the Carthaginians to strike coins.[21] Likewise, the 20,000 talent indemnity imposed by Sulla on the cities of Asia for supporting Mithradates was coined by Lucullus.[22]

Bullion for coinage came from friends as well as defeated enemies. Appian records that Brutus struck into coin treasures entrusted to him by a Thracian queen in 43 BC.[23] A gift (by bequest) from Ptolemy Alexander I of Egypt to the Roman people may account for the designation of issues of denarii in 86–85 BC as struck from *argentum publicum*.[24] A reasonably convincing case of a gift of bullion leading to the striking of a coinage can be seen from the inscription recording the gift of 15 talents of silver to Chios by Antiochus IV of Commagene on his honorific election to the chief magistracy there, and from the base silver coins inscribed 'gift of King Antiochus'.[25] An individual

[16] Cassiodorus, *Chronicon*, *sub anno* 158.

[17] G. Jones, 'The Roman mines at Riotinto', *JRS* 70 (1980), pp. 146–65, at pp. 161–3; the evidence for the decline in production of denarii under Commodus is taken by Jones from K. Hopkins, 'Taxes and trade in the Roman Empire (200 BC–AD 400)', *JRS* 70 (1980), pp. 101–25, at pp. 112–15. This decline is based on a decrease in the number of coins per year found in a variety of provinces. Since no check has been made for possible variations in survival rates for different periods by counting the dies used to produce the coins, the decline in production cannot be regarded as certain. For the debasements, see Walker, *Metrology*, vol. 3, pp. 125 ff; 138; 141.

[18] M. Price, 'Alexander's reform of the Macedonian regal coinage', *NC* 142 (1982), pp. 180–90, at p. 189; I. Carradice and M. Price, *Coinage in the Greek World* (1988), pp. 107–12.

[19] Diodorus xvi, 56, 5–7; cf. xvi, 36, 1.

[20] Xenophon, *Hellenica* vii, 4, 33 ff.; Kraay, *Archaic and Classical Greek Coins*, p. 106.

[21] C. M. Kraay, *Greek Coins and History* (1969), pp. 19–42; id., *NC*[7] 12 (1972), pp. 13–24; Price and Waggoner, *Asyut*, pp. 121–2; Diodorus xi, 26, 2–3.

[22] Plutarch, *Lucullus* iv, 1.

[23] Appian, *B.C.* iv, 75 (320).

[24] *RRC*, p. 605; E. Badian, 'The testament of Ptolemy Alexander', *RM* n.F 110 (1967), pp. 178–92.

[25] Howgego, *GIC*, pp. 86; 94.

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might give his own city a coinage; one of the clearest examples belongs to Paestum in the late Republic.[26]

The inventory given by Thucydides of Athenian treasures in 431 BC makes it clear that stored-up wealth, both coined and uncoined, could be viewed together as resources to spend.[27] An accumulation of miscellaneous coins could be recoined to make it easier to use, as by the Amphictions at Delphi in *c.* 338–333 BC.[28] Presumably in most cases wealth in the form of coin could simply be spent and we naturally hear more about the coining of other forms of treasure. Millar has gathered a number of examples of the melting of precious objects to provide bullion for coinage.[29] Otho, as *legatus* of Lusitania, came to support the proclamation of Galba against Nero bringing all the gold and silver that he had in the shape of drinking-cups and tables for conversion into coin.[30] The imperial treasures themselves appear not to have been so used but they were sold to raise cash at times of need. This was presumably because they were worth more as decorative objects than as bullion. Constantine is said by a number of sources to have used the statues and ornaments of the pagan temples to produce coinage.[31] They were no longer sacred; a similar fate (for a similar reason) had befallen the images of Domitian after his death.[32] In emergencies even sacred treasures could be coined. In 407/406 BC, when the Spartan occupation of Decelea had cut off Athens from its supply of silver, Athens resorted to striking coins from seven golden statues of Nike (an eighth survived).[33] In 297/296 BC Lachares coined the gold from shields on the Acropolis and from the chryselephantine statue of Athena in the Parthenon in order to pay mercenaries.[34] Alexander Zabinas (129–123 BC) seized the treasures of the Temple of Zeus at Antioch and carried off the golden Nike that stood in the palm of Zeus in order to pay troops.[35] The people rose against him when he made an attempt on the statue of Zeus itself. In 82 BC the Roman senate voted that the gold and silver ornaments of the temples be melted to pay the troops in the civil war against Sulla.[36] It was more likely the need to pay troops rather than greed which in the third century forced the emperor Maximinus to coin temple dedications,

[26] M. Crawford, 'Paestum and Rome. The form and function of a subsidiary coinage', *Supplement to AIIN* 18–19 (Rome, 1973), pp. 47–109, at pp. 53; 85 on no. 25/1 s(ua) p(ecunia) d(ono) d(edit), with a coining scene.

[27] Thucydides ii, 13, 3–5.

[28] P. Kinns, 'The Amphictionic coinage reconsidered', *NC* 143 (1983), pp. 1–22; E. J. P. Raven, 'The Amphictionic coinage of Delphi, 336–334 BC', *NC*[6] 10 (1950), pp. 1–22.

[29] Millar, *ERW*, pp. 144–53.

[30] Plutarch, *Galba*, 20, 3.

[31] Millar, *ERW*, p. 146.

[32] Dio lxviii, 1, 1.

[33] Schol. Aristophanes, *Frogs*, 720; E. S. G. Robinson, 'Some problems in the later fifth century coinage at Athens', *ANSMN* 9 (1960), pp. 1–15.

[34] *P. Oxy.* 17, 2082, fr. 4, lines 13–15; Pausanias i, 25, 7; J. Bingen, 'Le statère d'or attique de style pi V', in H. F. Mussche *et al.*, *Thorikos 1969* (Brussels, 1973), at pp. 16–18.

[35] Justin xxxix, 2, 5–6; Diodorus xxxiv, 28; W. Wroth, 'Greek coins acquired by the British Museum in 1896', *NC*[3] 17 (1897), pp. 93–113, at pp. 115–16.

[36] Valerius Maximus vii, 6, 4.

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statues of the gods, honorary presentations to the heroes and any ornamentation on public buildings or city decorations.[37] This kind of sacrilege was a horror normally associated with war: if the gods are plundered and temples burned 'Numina miscebit castrensis flamma monetae'.[38]

Purchase was another possibility, if a state had reserves in one metal but preferred coinage in another (for example because of the scale of payments to be made or the greater acceptability to the recipients of one metal rather than another). We hear of forced purchases of gold in Egypt from AD 300,[39] perhaps for coinage, and in cases where mines were exploited by individuals under concession the state could presumably purchase bullion over and above that due to it under the concession agreements.[40] A further possibility arises at times when states coined on behalf of individuals for a fee, in which case the state did not even need to own the bullion in order to produce coin.[41]

It should be clear from this brief review of the sources of bullion that there are advantages in considering coinages from the point of view not only of the use for which they were intended but also of the source of metal which made them possible. Coinages which derived from gifts, windfalls and booty should not necessarily be interpreted in the same way as those which arose from a pressing need to make payments. The availability or not of bullion could affect patterns of coin production every bit as much as could variations in requirements for state expenditure. That said, the need to strike coin to make payments was paramount in some cases, notably in the instances of seizure of sacred treasures cited above. We now turn to examine the nature of the expenditures which could give rise to coinage.

EXPENDITURE AND NEW COINAGE

As stated above, we do not have the evidence to show a systematic correlation between coinage output and military expenditure under the Roman Republic. Two things follow. First, there is no basis for the notion that expenditure was regularly paid only in new coin under the Republic.[42] The second corollary is that Crawford's quantitative proof that military

[37] Herodian vii, 3, 5 cf. vii, 3, 3 (pay for the troops).

[38] Lucan, *De Bello Civili* i, 379–80.

[39] T. C. Skeat, *Papyri from Panoplis* (Dublin, 1964), papyrus 2, lines 215–21 (pp. 90–3); Millar, *ERW*, p. 152. Millar states that the consignment of gold sent from Oxyrhyncus to Nicomedia in 304 did not coincide with a significant issue of aurei from Nicomedia at that time. In fact *RIC* vol. VI, p. 544, argues that gold was struck in some quantity from *c.* 303 at any rate to 304. In any case the gold could have been sent to another mint for striking or held in store.

[40] For the organisation of the exploitation of mines see J. F. Healy, *Mining and Metallurgy in the Greek and Roman World* (1978), Chapter 5; R. Osborne, *Demos: The Discovery of Classical Attika* (Cambridge, 1985), chapter 6; J. S. Richardson, 'The Spanish mines and the development of provincial taxation in the Second Century BC', *JRS* 66 (1976), pp. 139–52.

[41] See below.

[42] *Pace RRC*, pp. 617–18.

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expenditure dominated the state budget (because it dictated coin production) is shown not to be valid. Nevertheless, the high level of military spending, even if it cannot be quantified accurately, suggests that the requirements of armies (and navies) were an important motivation for coinage.[43] It is easiest to identify coinages connected with periods of warfare, although it should not be forgotten that standing armies required regular pay and often substantial sums on discharge. Kraay has set out some instances of Greek coinages which are to be connected with war and Roman examples are not hard to find.[44]

How are we to recognise military coinages? Some are obligingly inscribed: a group of Siculo-Punic coins is marked *Machanat* (the camp), the word καταστρατεία (on military expedition?) occurs on Parthian coins,[45] the Pompeians in Africa struck an issue signed by a *leg(atus) f(isci) c(astrensis)* (if this is the correct resolution).[46] The typology is sometimes so specifically military that the inference of a military function for the coinage seems reasonable. So the coins of Tarsus under local dynasts down to *c.* 400 BC show the Persian soldiers for whose pay (it is argued) the coinage was struck, but from the early fourth century the coins of the satraps show the Greek hoplites who served with the Persians after the end of the Peloponnesian War. The legionary aurei and denarii of M. Antonius provide a Roman example.[47] The context in which some coinages were issued indicates beyond doubt their military function: for example those struck by (or for) the Libyan mercenaries at the end of the First Punic War,[48] Hannibal's issues in Bruttium,[49] Rome's massive coinage in 90 BC during the Social War (larger than in any other year of the Republic),[50] or Roman denarii produced in north Italy and Spain during the war against Sertorius.[51] Sometimes a monetary phenomenon demands a military explanation. The spectacular flowering of mints striking pegasi in north-west Greece and in Sicily *c.* 340 BC, and the fact that the bulk of the output from all mints reached Sicily, cannot be dissociated from the Corinthian expedition under Timoleon to support the Greek cities of Sicily against the Carthaginians. Under Nero the silver coinages of the Roman East obediently followed Corbulo on

[43] On the dominance of the army in Roman state expenditure see the passages cited by Crawford, *ANRW*, II.2, pp. 561–2, n. 4; K. Hopkins, 'Taxes and trade in the Roman empire (200 B.C.—A.D. 400)', *JRS* 70 (1980), pp. 101–25, at pp. 116–17.

[44] C. M. Kraay, 'Greek coinage and war', in *Ancient Coins of the Graeco-Roman World, The Nickle Numismatic Papers* (edited by W. Heckel and R. Sullivan) (Waterloo, Ontario, 1984), pp. 3–18, where the evidence for the Greek examples may be found.

[45] *BMC Parthia*, p. 40.

[46] *RRC*, no. 461.

[47] *RRC*, no. 544.

[48] I. Carradice and S. La Niece, 'The Libyan War and coinage: a new hoard and the evidence of metal analysis', *NC* 148 (1988), pp. 33–52.

[49] M. Crawford, *CMRR*, p. 67; E. S. G. Robinson, 'Carthaginian and other south Italian coinages of the Second Punic War', *NC*[7] 4 (1964), pp. 37–64.

[50] Crawford, *RRC*, no. 340.

[51] Crawford, *RRC*, nos. 366; 393, with p. 82.

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campaign.[52] Perhaps clearest of all are the cases in which historians record that coinages were for military purposes. We have already seen the temple treasures of Athens, Olympia, Delphi and Antioch used to provide coin for troops; sacrilege arose from compelling needs and demanded comment. Faced with a lack of silver on his expedition against Olynthus in 363–359 BC, Timotheos devised a token bronze coinage in order to pay his troops.[53] Casual references in Sallust and Plutarch record that coinages were struck by Metellus for the Spanish War and by Lucullus for the war against Mithradates.[54] There is no need to multiply examples. The danger is that concentration on military expenditure causes us to lose sight of other types of expenditure.

Food must have been an important item. That expenditure on food could evoke a coinage can be nicely illustrated by the denarii struck by the quaestors Piso and Caepio in *c.* 100 BC specifically for the purchase of corn (*ad fru(mentum) emu(ndum)*).[55] The reverse type showing the two magistrates seated on their bench between ears of corn is echoed by an issue under two plebeian aediles in *c.* 86 BC, but without the helpful inscription.[56] It must remain uncertain whether an issue such as the latter was struck specifically to meet the type of expenditure shown and the uncertainty is even greater in other cases where we cannot even be sure of the status of the individuals who signed the coinages. Crawford prefers to view all individuals for whom no magistracy is specified as moneyers. He thus interprets 'aedilician' types (types referring to games in the circus and to corn distributions, both of which lay within the sphere of duties of the aediles) as electoral promises by moneyers.[57] This need not be so and the more natural interpretation of 'aedilician' types as struck by aediles (or magistrates with similar functions) remains possible, at least for some cases.

Expenditure connected with games may lie behind the issues by curule aediles which make reference to their responsibility for the *Ludi Megalenses*, and behind a cluster of types in *c.* 78 BC which make reference to the *Ludi Romani, Plebeii, Cereales, Megalenses and Apollinares.*[58] We are on firmer ground in associating the substantial issue under the curule aediles M. Aemilius Scaurus and P. Hypsaeus in 58 BC with the memorably

[52] Walker, *Metrology*, vol. 3, p. 113.

[53] ps. Aristotle, *Oeconomica* ii, 1350a, 1; E. S. G. Robinson and M. J. Price, 'An emergency coinage of Timotheos', *NC*[7] 7 (1967), pp. 1–6; the identification of the issue has been rendered doubtful by recent finds: M. J. Price, 'The function of early Greek bronze coinage', in *Atti del VI Convegno del Centro Internazionale di Studi Numismatici – Napoli, 17–22 Aprile 1977* (Rome, 1979), pp. 351–8, at p. 358, n. 19.

[54] Sallust, *Historiae* ii, 34 (ed. Maurenbrecher); Plutarch, *Lucullus* 2, 2.

[55] *RRC*, no. 330, with p. 73: H. B. Mattingly, *NC* 137 (1977), p. 205.

[56] *RRC*, no. 351.

[57] *RRC*, pp. 601; 729, with n. 3.

[58] *Ludi Megalenses*: *RRC*, nos. 356; 409/2; 431. Issue of 78 BC: *RRC*, no. 385; Th. Mommsen, *Geschichte des römischen Münzwesens* (Berlin, 1860), pp. 620–1; note the possible expansion of SC DT as *senatus consulto de thesauro*.

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extravagant aedilician games of the former.[59] A Greek example of a coinage connected with games might tentatively be sought in the coins in the name of Elis presumed to have been struck at Olympia, although we actually know nothing of the occasions or organization of that coinage.[60] Under the Roman Empire games were an important element in the typology of the civic coins, although agonistic types were not confined to the period of games.[61]

Distributions of money are recorded from early times (Croesus gave two gold staters to every citizen of Delphi)[62] but the best evidence for a link with new coinage comes from the Later Roman Empire. Quinquennial *vota*, which provided occasions for military donatives, made a substantial impact on the typology of the coins. Unfortunately, the *vota* types were not confined to the years in which donatives were made – for example the tricennial *vota* of Constantius II belonged in AD 353 but the corresponding type was still being used when Julian was made Caesar in 355 and Augustus in 360 – so one can never be certain from the types of the coins which *vota* issues were struck specifically for distribution.[63] However, Ammianus records that when Procopius claimed the purple under Valentinian he distributed gold coins minted with his image (unless these were bribes rather than distributions), the anonymous *De Rebus Bellicis* refers to bronze being struck *ad dona militaria* and Claudian mentions gold coinage engraved for a donative after *vota*.[64] From the early imperial period the issue at Chios inscribed 'gift of King Antiochus' provides a plausible example of a coinage struck for distribution in a non-military context, as may other formulae used on civic coins under the Roman Empire.[65]

It is highly likely that public works, such as the great building programmes on the Athenian Acropolis, and pay for public servants (jurors and councillors were paid at Athens from the 450s BC) gave rise to coinages, but we cannot now securely identify them.[66] Crawford chose to ignore the costs of central administration under the Roman Republic but the fact that we have no way of estimating these costs does not mean that they were insignificant.[67] Settlements also cost money: issues of quinarii can be

[59] *RRC*, no. 422; Pliny, *HN* xxxvi, 24 (113–15); H. B. Mattingly, *NC* 137 (1977), pp. 212–13, notes that the reverse type of Jupiter in a quadriga refers to the *Ludi Romani*.

[60] Kraay, *Archaic and Classical Greek Coins*, pp. 103–7.

[61] See R. Ziegler, *Städtisches Prestige und Kaiserliche Politik. Studien zum Festwesen im 2. und 3. Jahrhundert n. Chr.* (Düsseldorf, 1985); Howgego, *GIC*, p. 89.

[62] Herodotus i, 54.

[63] E.g. *RIC*, vol. 8, pp. 218–27; P. Bastien, *Monnaie et Donativa au Bas-Empire* (Wetteren, 1988), contains much useful information but does not get to grips with this problem.

[64] Ammianus xxvi, 7, 10–11; *De Rebus Bellicis* i, 9; Claudian, *In Rufinum* ii, 339–42.

[65] Howgego, *GIC*, pp. 94–5.

[66] For a discussion of these aspects of expenditure in the context of coinage see N. K. Rutter, 'Early Greek coinage and the influence of the Athenian state' in *Coinage and Society in Britain and Gaul: Some Current Problems*, edited by B. Cunliffe (1981), pp. 1–9, at p. 5.

[67] *RRC*, p. 697.

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plausibly linked to occasions of expenditure in Cisalpine Gaul, where this denomination was preferred, and those in *c.* 101 BC and *c.* 99–97 BC have been associated by Crawford with colonising activity and viritane settlement of veterans.[68] An issue is certainly to be associated with the foundation of the *colonia* at Narbo in 118 BC (it appears to be signed by the *IIviri col. deduc.*, although their title is not given), but we do not know whether it was intended to meet expenditure connected with the foundation or to provide a coinage for the *colonia.*[69] Coinage might also be produced to meet external expenditure of one kind or another. Philip of Macedon struck gold not only to raise an army but also to bribe Greeks.[70] Herodotus disparaged a story that Polykrates struck lead coins plated with gold to bribe the Spartans to raise the seige of Samos in 525/524 BC; we may share his scepticism but also be entertained by the thought that such a coinage does exist.[71] Ransoms and subsidies to barbarians could be important at certain periods but we do not usually know whether payments were made in old or new coin or bullion.[72]

Thus far a variety of expenditures which might call for the production of a coinage have been noted – military requirements, food, games, distributions, public works, public servants, settlements and external payments – and most have been illustrated by examples. One might think of other kinds of expenditure, such as the costs of an imperial visit under the Roman Empire, but these are probably the principal ones. The link between the need to make payments and the striking of coinage was clearly an important one in some cases but the validity of expenditure as a general explanation for coinage is weakened by consideration first of occasions on which expenditure did not call forth new coinage and then of reasons other than expenditure for producing coins.

EXPENDITURE WITHOUT NEW COINAGE

The major problem in connecting coinage with expenditure is that we hardly ever know to what extent existing rather than new coinage was used to make payments.[73] Without that knowledge it is impossible to reconstruct a state budget from an estimate of annual coinage production. In some cases the use of foreign coins could be an alternative. Such problems are easily seen from times of known expenditure which produced little or no corresponding new coinage. Sparta, for example, had no coinage of its own for making external payments until the third century BC, and Corinth produced little

[68] Crawford, *CMRR*, pp. 182–3.

[69] *RRC*, no. 282.

[70] Diodorus xvi, 8, 6–7.

[71] Herodotus iii, 56, 2; J. P. Barron, *The Silver Coins of Samos* (1966), p. 17; Kraay, *Archaic and Classical Greek Coins*, pp. 29–30.

[72] See, for example, M. F. Hendy, *Studies in the Byzantine Monetary Economy c. 300–1450* (Cambridge, 1985), pp. 260 ff.

[73] On stores of coin under the Roman Empire see Millar, *ERW*, pp. 144–53.

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coinage at least during the earlier part of the Peloponnesian War.[74] Some use of old coin and foreign coin was thus inevitable and can be documented.

A few Greek accounts preserved on stone allow us glimpses of the use of foreign coin.[75] Contributions from states and individuals to a Spartan war-fund, probably during the Archidamian War, included Persian gold darics from Ephesus and Aeginetan staters from friendly Chians. Such coinages took on an 'international' flavour and were perhaps particularly suitable for payments between states. Thus Byzantium contributed 584 Lampsacene gold staters and 16 Attic silver drachms to the Boeotians for the conduct of the Sacred War in 355–351 BC. But foreign coins could be used for internal payments too, even by states which produced substantial coinages of their own. This can be seen from the allocation by the Athenians of a sum of Cyzicene gold staters to their own trierarchs for operations off Argos in 418/417 BC.

By contrast, one would not expect old coin to be differentiated from new in inscriptions and so it is necessary to resort to other kinds of evidence and argument. The use of old coin to make payments under the Roman Empire is likely to be the explanation of the contents of a hoard of 1,127 silver coins found in a small military outpost at Lay in the Loire.[76] The coins dated from M. Antonius to Severus Alexander and were almost all in mint condition. They were arranged in the shape of a cylinder (like a saucisson according to the finder) and are likely to have been originally contained in a wooden box. The most probable explanation of this arrangement, the condition of the coins, and the provenance is that the parcel was a consignment of pay for the troops. Furthermore, detailed work on the level of production of coinage under Domitian has shown that new coinage could have accounted for only a small proportion of annual state expenditure.[77] Coin in circulation at all times flowed back to the treasury through taxes, confiscation, booty, bequest and the like and we are in no position to make assumptions about how far it was recoined before being spent again (except perhaps in the later Roman period).

A further complication is that expenditure did not need to be in coin at all.

[74] Sparta: S. Grunauer-von Hoerschelmann, *Die Münzprägung der Lakedaimonier* (Berlin, 1978). Corinth: H. B. Mattingly in his review of Carradice and Price, *Coinage in the Greek World*, for *NC* 149 (1989), pp. 228–32, has re-examined this question and suggested that Corinth may have had more coinage during the war than was previously thought. There remains remarkably little coinage which can be attributed to Corinth during the Archidamian War. The possible collective use of the coinage of Sicyon by Sparta and her allies (including Corinth) during the Peloponnesian War seems likely to remain a matter for debate.

[75] Spartan war-fund: Meiggs and Lewis, 67; contributions to Sacred War: M. N. Tod, *GHI*, vol. 2, no. 160; Athens in 418/417 BC: Meiggs and Lewis, 77, lines 13–15.

[76] *Revue archeologique de l'est et du centre-est* 1 (1950), pp. 175–80.

[77] A. M. Burnett, *Coinage in the Roman World* (1987), p. 95, drawing on the work of I. Carradice, *Coinage and Finances in the Reign of Domitian, AD 81–96, BAR International Series 178* (Oxford, 1983).

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The use of credit, as by Rome during the Second Punic War,[78] was exceptional, but it is much harder to assess the importance of payments in bullion. The pseudo-Aristotelian *Oeconomica* states that it was within the scope of the royal economy to decide whether expenses were to be paid for in currency or in goods of equivalent value.[79] Book II of the *Oeconomica* was composed after the death of Alexander and the ideas in this passage have clearly been influenced by the structure of the Persian Empire. Payment in goods rather than currency should not, however, be dismissed as a purely oriental option. The possibility of making payments in bullion rather than coin was open to the Romans too.

We know that the Roman state made some payments in silver bullion but the possibilities can be more easily explored in respect of gold.[80] Rome struck no gold coinage between the Second Punic War and the time of Sulla, and no substantial issues until 46 BC.[81] Yet gold in the form of booty poured in; notices in Livy relating to triumphs and ovations record an inflow of some 38,000 pounds of gold between 200 and 167 BC.[82] It came not only as bullion but as vases, plate, crowns, torques and the like, or coined in Philippei.[83] Gold booty continued to arrive but cannot be quantified after the narrative of Livy gives out. Indemnities imposed on defeated enemies were specified in silver rather than gold, a fact which surprised the Elder Pliny, but the Aetolians at least were allowed to pay in gold.[84] Gold will also have been derived during the course of the second century BC from the mines of Spain, Italy, Macedon (after 158 BC) and from an exceptional find in Noricum.[85] In 157 BC the *aerarium* contained 17,410 pounds of gold, worth over four times as much as the combined holdings of silver bullion and silver coin.[86] It is

[78] T. Frank, *An Economic Survey of Ancient Rome, vol. 1, Rome and Italy of the Republic* (Baltimore, 1933), pp. 84–8; Crawford, *CMRR*, pp. 60–2.

[79] ps.-Aristotle, *Oeconomica* ii, 1345b.

[80] Silver bullion: Crawford, *CMRR*, p. 143 citing in particular the allocation of 150,000 pounds of silver to the Ligurians transferred to Samnium (Livy xl, 38, 6) and the gift by Paullus of five pounds of silver out of the booty of Pydna to Q. Aelius Tubero (Valerius Maximus iv, 4, 9).

[81] The issues from the Second Punic War to 46 BC are *RRC*, nos. 359; 367; 375; 381; 402; 460; 466.

[82] Frank, *An Economic Survey of Ancient Rome*, vol. 1, p. 138; in respect of the triumph over Perseus in 167 BC Livy gives only a total figure; the proportion which was in gold has been derived from Diodorus xxxi, 8, 11 and Plutarch, *Aemilius Paulus* 32–3. It is difficult to reproduce exactly Frank's calculations (which in any case depend on his stated assumptions) but they are of the right order.

[83] The frequent mention of *philippei* in the plays of Plautus cannot be taken as evidence of their use at Rome because they reflect the Greek origin of the genre. References in G. Lodge, *Lexicon Plautinum* (Leipzig, 1926–1933), vol. 2, pp. 325–6.

[84] Pliny, *HN* xxxiii, 15 (51); Livy xxxviii, 11, 8–9.

[85] Frank, *An Economic Survey of Ancient Rome*, vol. 1, pp. 256; 262–3; Noricum: Strabo iv, 6, 12, quoting Polybius.

[86] Pliny, *HN* xxxiii, 17 (55); the calculation assumes that denarii were struck at 84 to the pound and a gold-silver ratio of 10:1, but any plausible ratio supports the point that the gold was worth considerably more than the silver. The calculations by Crawford, *RRC*, p. 635 are

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reasonable to question whether all this gold was systematically alienated (spent?) to acquire bronze and silver for coining.[87] Some may have been used for decorative purposes but there remains the distinct possibility that the Roman state was accustomed to make substantial payments in gold bullion. We happen to know that the murderer of Gaius Gracchus was rewarded with gold in 121 BC.[88] We may speculate that payments in gold bullion continued at least until the substantial issues of 46 BC, as gold coinages were small and infrequent from 84/83 BC until 46 BC. Sulla took some 9,000 pounds of gold from the property of temples for his equipment for the Mithradatic War in 88 BC.[89] The senate decreed that gold and silver temple ornaments be melted down to meet military expenses against Sulla in 82.[90] The younger Marius had had 14,000 pounds of gold conveyed from the Capitol after it burned in 83.[91] This gold was returned to Rome in 81 in the triumphal processions of Sulla, in which was also carried 15,000 pounds of gold from Sulla's own victories (a small fraction of the sums he had obtained from Asia).[92] The vast booty of Pompey's triumphs naturally included gold.[93] Caesar removed 3,000 pounds of gold from the Capitol during his first consulship in 59 and brought back so much gold from his Gallic victories that the price fell.[94] In 49, Caesar was able to obtain 4,135 pounds of gold from the *aerarium sanctius*, the revenue from the 5% tax on manumissions which was set aside for emergencies.[95] The gold in the *aerarium sanctius* was kept in the form of bars (*lateres*), as may have been the other bullion stored in the *aerarium*.[96] The monetary use of gold bars may be reflected by their presence in two hoards of denarii, although the occurrence of bars in the hoard of 49 BC may be seen as an exceptional way of circumventing the restrictions on the hoarding of coin revived by Caesar.[97]

The use of bullion to make state payments under the Republic is thus plausible, and there is also some evidence for the practice in the later Roman period. For example, a letter of AD 300 preserved on papyrus contains instructions from the *procurator* of the lower Thebaid to the *strategos* of the nome of Panoplis to deliver fifty pounds of uncoined silver and four folles to

incorrect as they employ a rate of 120 denarii to the pound (despite his note that he is using 84 to the pound).

[87] Compare Crawford, *CMRR*, p. 143: 'Rome presumably alienated large quantities of gold and silver bullion in exchange for bronze for coinage.'

[88] Pliny, *HN* xxxiii, 14 (48); Plutarch, *Gaius Gracchus* 17, 3–4.

[89] Appian, *Mith.* 22.

[90] Valerius Maximus vii, 6, 4.

[91] Pliny, *HN* xxxiii, 5 (16–17).

[92] Frank, *An Economic Survey of Ancient Rome*, vol. 1, p. 232.

[93] Plutarch, *Pompey* 45.

[94] Suetonius, *Caesar* 54, 2–3.

[95] Orosius, *Historiarum adversum Paganos* vi, 15, 5; for the tax: Livy vii, 16, 7; for the use of the gold in 209 BC: Livy xxvii, 10, 11.

[96] Pliny, *HN* xxxiii, 17 (56); cf. Nonius Marcellus, *De Conpendiosa Doctrina* 12 (Lindsay, p. 837); Varro *de Vita Populi Romani*, lib. iii: 'nam lateres argentei atque aurei primum conflati atque in aerarium conditi'.

[97] *RRCH*, nos. 259 (cf. 307); 357; restrictions on hoarding: Dio xli, 38, 1 ff.

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the *praepositus* of the *lancearii* of the *legio III Diocletiana*.[98] It is uncertain whether the silver was intended for pay or donatives or both, but the use of silver bullion for donatives seems to be attested by Julian's promise of 5 solidi and a pound of silver to each soldier in AD 360.[99] In this case it remains possible that the pounds of silver were distributed in the form of coin but silver ingots which may have been used for such payments survive from the fourth century.[100]

It is thus clear from the use of old or foreign coin, credit, or bullion that expenditure did not automatically require new coin. It is equally clear that not all issues of coinage were motivated by the need to make expenditure.

NEW COINAGE WITHOUT EXPENDITURE

One obvious occasion for a new issue which need not have been connected with expenditure was a coinage reform. This might be provoked by a need to raise money. Book II of the pseudo-Aristotelian *Oeconomica*, which is full of tips on devious ways to generate funds, has two delightful stories about coinage reform.[101] The citizens of Syracuse demanded back the money which Dionysius had borrowed from them, whereupon he commanded them to bring him whatever money they possessed under threat of death. He took this new silver, re-struck the drachms as didrachms, and paid them back in full. (Polyaenus has a similar story from the Bosporan Kingdom in which Leukon recalled the coinage in circulation, re-struck it with double the face value, and gave half back.[102]) At Athens Hippias is said to have declared the existing coinage invalid and to have fixed a (presumably lower) price for it. He then recalled the coinage and reissued it again without even bothering to re-strike it. We do not need to believe in the historicity of these incidents in order to take the point. Revaluations could raise money. Under Hadrian the cistophori in circulation in the province of Asia were re-struck at about twenty mints and it has been plausibly argued that at this time the cistophoric tetradrachm was revalued from three to four denarii.[103] Lowering the standard of a coinage, whether by lowering its weight or debasing the metal, provided a source of money, if coinage of the former standard was utilized for bullion. The massive recoinage at Alexandria in Egypt under Nero may have extracted considerable amounts of silver from the earlier and finer coinage at the time of financial crisis following the great fire at Rome in July

[98] T. C. Skeat, *Papyri from Panoplis* (Dublin, 1964), papyrus 2, lines 298–304 (pp. 104–7).
[99] Ammianus xx, 4, 18.
[100] P. Bastien, *Monnaie et Donativa au Bas-Empire* (Wetteren, 1988), p. 21, n. 3 (pl. 1).
[101] ps.-Aristotle, *Oeconomica* ii, 1349b (Dionysius); 1347a (Hippias).
[102] Polyaenus vi, 9, 1.
[103] Walker, *Metrology*, vol. 2, p. 64; vol. 3, pp. 72–3; Howgego, *GIC*, p. 53; W. Metcalf, *The Cistophori of Hadrian*, Numismatic Studies 15 (New York, 1980).

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AD 64.[104] Since the new coins contained less silver than those being replaced, it was possible either to strike more coins than were being withdrawn or to strike the same number and extract the surplus silver. Trajan's debasement of AD 107 was accompanied by the recall of old coin and various old types were 'restored' (*rest(ituit)*), giving the flavour of what had been melted.[105] The connection of coinage with coinage reform rather than with patterns of expenditure can be well illustrated by the enormous issues which follow the reduction in the silver standard in Syria in AD 108–9 but fail to reflect Trajan's Parthian War.[106]

The imposition and maintenance of a closed currency system was another cause of re-coinage. This is best seen in practice in a Ptolemaic papyrus of 258 BC, from which it is clear that only Ptolemaic gold was current in Egypt. The mint master in the normal course of events re-struck gold for 'the foreigners who come here by sea, the merchants, the forwarding agents, and others'. The currency system of Egypt remained closed until Diocletian and presumably continued to be a cause of re-coinage.[107] A closed currency system also operated from the middle of the fourth century at Olbia on the Black Sea, where all sales and purchases had to be carried out with the local bronze and silver coinage.[108] It has been plausibly argued that a closed currency system operated also in Byzantium and Chalcedon *c.* 235–220 BC and in the Attalid kingdom from *c.* 175 BC, and that an attempt was made to impose one in Syria *c.* 173/172 BC.[109] The countermarking of foreign coins in relation to these episodes was in effect a re-coinage.

A re-coinage might also be required to standardize a miscellaneous group of coins or to renew a worn coinage. The Amphictions at Delphi melted and struck into their own name the varied coinages in their treasury in *c.* 338–333 BC; the treasurers' accounts record the sums re-coined.[110] There was clearly an advantage in standardization prior to making payments. Worn coin also presented a problem; it was probably normal to require that worn coins be accepted at full face value but difficulties arose in practice. A

[104] E. Christiansen, 'On denarii and other coin-terms in the papyri', *ZPE* 54 (1984), pp. 271–99; id., 'The Roman coins of Alexandria (30 BC to AD 296). An inventory of hoards', *Coin Hoards* 7 (1985), pp. 77–140; id., *The Roman Coins of Alexandria. Quantitative Studies. Nero, Trajan, Septimius Severus* (Aarhus, 1988), p. 308; but on the unreliability of the estimates of coin production see the review by Howgego, *JRS* forthcoming.

[105] *RIC*, vol. 2, pp. 302–13; Dio lxviii, 15 (Xiphilinus); Walker, *Metrology*, vol. 2, pp. 54–7; vol. 3, pp. 121–2.

[106] Walker, *Metrology*, vol. 2, pp. 103–4; vol. 3, pp. 122–4.

[107] *P. Cairo Zen.* 59021; *Sel. Pap.* vol. 2, 409; translated by Austin, *The Hellenistic World*, pp. 410–11, no. 238; the closed system seems to go back to Ptolemy I: O. Mørkholm, 'Some reflections on the production and use of coinage in Ancient Greece', *Historia* 31 (1982), pp. 290–305, at p. 298; on the closed currency system under Rome, see n. 104.

[108] Dittenberger, *Sylloge*[3], no. 218; translated by Austin and Vidal-Naquet, *ESH*, pp. 330–3, no. 103.

[109] Mørkholm, *Historia* 31 (1982), pp. 298–305.

[110] See n. 28 above.

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rescript of Hadrianic date to Pergamum, which is concerned with various malpractices perpetrated by moneychangers and by the public in relation to the exchange of silver and bronze coin, defines *aspratoura* as an abuse but is evidence of the practice.[111] The precise nature of *aspratoura* is unclear but the abuse probably lay in the demanding of new coin or the discounting of worn ones. The word might have applied to the rapacious demand by Nero for payment of taxes in fresh coin (*nummum asperum*).[112] Similarly, in the late fourth or early third century BC the city of Arcesine on Amorgos demanded the repayment of loans in current coin (δόκιμον) of full weight (ὁλοσχε[ρε]ς).[113] The problem surfaces in casual references such as Martial's description of a poor man with a worn denarius or Theophrastus' characterization of the boorish farmer who refuses coins because they are of too poor quality.[114] The Ptolemaic papyrus of 258 BC mentioned above is clear on the matter.[115] A bureaucratic muddle had halted work at the mint: 'as for the people in the city (Alexandria) they are all reluctant to use worn gold coins. For none of them know to whom he can refer after adding a little to get back fine gold or silver in exchange'. It is hard to believe that the need to renew a worn currency was not a significant factor in some decisions to coin throughout antiquity. To take a later example, the majority of cistophori had been in circulation for over a century when they were re-struck under Hadrian.[116] That the motivation was to renew the currency rather than simply to facilitate expenditure of old coin in the provincial *fiscus* can be seen from the decision to strike at about twenty different mints (the coins were dispersed in circulation). The renewal of the coinage is explicit in the restored issues of the Roman principate. The formula *restituit* is not found on coins until the reign of Titus but the deliberate imitation of earlier types under Vespasian suggests that the idea goes back earlier.[117]

A further technical matter which might (but need not) call into question the model of the production of coin purely for expenditure is the operation of coinage alliances. The best known example is the agreement between Phocaea and Mytilene to produce electrum in alternate years.[118] The commitment to produce coinage in alternate years might have involved some production which was not directly connected with expenditure. The terms of other alliances could, in principle, have affected patterns of production. However, although many other alliances can be demonstrated from the coins themselves, it is only for Phocaea and Mytilene that any terms are recorded.

Less technical is the desire to make a profit. One result of the hitch in the

[111] *OGIS* 484, line 25; Howgego, *GIC*, p. 10, n. 78.
[112] Suetonius, *Nero* 44, 2.
[113] *IG* XII (7) 69, 21–3.
[114] Martial, *Epigrammata* ii, 51, 1–2; Theophrastus, *Characters* 4, 11.
[115] See n. 107.
[116] See n. 103.
[117] T. V. Buttrey, 'Vespasian as moneyer', *NC*[7] 12 (1972), pp. 89–109.
[118] M. N. Tod, *GHI*, vol. 2, no. 112; J. F. Healy, 'Notes on the monetary union between Mytilene and Phokaia', *JHS* 77 (1957), pp. 267–8.

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Ptolemaic mint in 258 BC was that 'the king's revenues are suffering no small loss'.[119] Further evidence is to be found in an inscription from Sestos which belongs after the end of the Attalid kingdom. The inscription honours a certain Menas for his generosity and services to the city, one of which was the undertaking of an appointment to provide the city with a bronze coinage. The inscription is explicit about why the people decided to use its own bronze coinage. The second reason given was that the people should receive the profit resulting from this source of revenue (the first reason was that the city's coin type should be current).[120] Profit could derive from the overvaluation of the coinage, from selling the monopoly of exchange, or from charging a fee for striking coins for others.

Overvaluation for these purposes is the excess in the face value of a coin over the value of the bullion, after deducting the cost of minting (*c.* 5% in some cases), including any loss of bullion during the refining and minting process.[121] In no case can this be quantified exactly. Not only are the actual costs of minting hardly ever known but we also need to be sure that we are using the right figures for the cost of bullion. Minimum figures for overvaluation may be estimated in cases where contemporary coinages had the same face value but different metal contents. For example cistophoric tetradrachms passed within the Attalid kingdom at the same value as Attic tetradrachms, as cistophoric countermarks on Attic tetradrachms show, but the cistophoric tetradrachm weighed only three Attic drachms.[122] It was thus overvalued by a minimum of 33% (one Attic drachm per cistophoric tetradrachm); the overvaluation of the Attic tetradrachms is not known. In a similar way the silver coinages of the Roman East were systematically overvalued against Roman denarii.[123] If one ignores minting costs one can derive some idea of overvaluation from prices for bullion expressed in coin, as long as one knows the original metallic content of the coins. Calculations based on the price of uncoined silver at Coptos in Egypt in *c.* AD 117 have been quite widely used to show that the Egyptian tetradrachm only contained about 60% of the silver it could buy (i.e. was overvalued by about 67%), if no account is taken of minting costs.[124] We know, however, that the

[119] See n. 107.

[120] *OGIS* 339; translated by Austin, *The Hellenistic World*, pp. 348–52, no. 215.

[121] Mørkholm, *Historia* 31 (1982), pp. 290–6; the accounts of the Amphictions at Delphi listed the difference between the amounts of coin sent for re-minting and the sums received back (ἀπουσία). On overvaluation, see also Howgego, *GIC*, pp. 68–9.

[122] Mørkholm, *Historia* 31 (1982), pp. 300–1.

[123] Walker, *Metrology*, *passim*.

[124] *P. Giess.* 47, lines 28–9, as restored in F. Preisigke, *Berichtigungsliste der griechischen Papyrusurkunden aus Ägypten* (Berlin and Leipzig, 1922), p. 170. For the use to estimate overvaluation see, for example, J. Guey, 'De "L'or des Daces" (1924) au livre de Sture Bolin (1958)', in *Mélanges d'archéologie, d'épigraphie et d'histoire offerts à Jérôme Carcopino* (1966), pp. 445–75, at pp. 449–51; T. V. Buttrey, *AJA* 65 (1961), 84, calculating that bullion was worth 56·7% of coin. For the context in which the price belonged: L. West and A. Johnson, *Currency in Roman and Byzantine Egypt* (Princeton, 1944), pp. 93–4.

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price of bullion was fluctuating daily, and unrest in Egypt at this date may have led to a preference for ready coin. For our purposes it is the bullion price at the time and place of minting which is relevant. The least problematic calculations may be derived from the prices for bullion given in Diocletian's Prices Edict and from the imperial letter concerning a revaluation of the coinage which accompanied it.[125] Gold was not overvalued; it was worth the same in coined and uncoined form. Silver as bullion was valued at 6,000 denarii a pound but 96 argentei of 100 denarii each could be struck from a pound; in other words silver coin was overvalued by 60%. The case of the billon nummus of *c.* 10 g. is problematic but, if it is correct to restore the imperial letter so that it records a doubling in the face value of a nummus to 25 denarii and to accept on the basis of the XXI mark that it had a theoretical composition of twenty parts bronze to one part silver (in practice it seems lower), then the nummus was overvalued in AD 301 by about 185%. For our purposes precision is less important than the demonstration that overvaluation could provide an enticement to coin.

A category of coinage which not only was potentially profitable but clearly had no connection with state expenditure was that struck by the state on behalf of individuals for a fee. This practice was a feature of the medieval period;[126] evidence for it in the ancient world is patchy but may have been underestimated. Again the best illustration comes from the Ptolemaic papyrus of 258 BC.[127] For a fee 'the foreigners who came here by sea, the merchants, the forwarding agents and others' could get their foreign gold coins struck into Ptolemaic gold and the people in the city could have their worn coins re-struck. The possibility of getting worn coins re-struck shows that minting for individuals was not simply a practice designed to enable a closed currency system to be enforced, although this will have been the explanation of the minting fee specified in the fifth century Athenian Coinage Decree, which sought to enforce the exclusive use of Athenian coin by the allies.[128]

It has become customary to reject the possibility of minting for individuals at Rome before the fourth century AD. This may be right but there is one piece of evidence which is worth bearing in mind. In 49 BC, when a shortage of coin meant that no one was paying their debts, Cicero wrote to Philotimus, his wife's freedman and steward, instructing him to get his travelling money from the Mint (*a Moneta*) or from Atticus' banking friends, the Oppii.[129]

[125] A clear summary may be found in Hendy, *Studies in the Byzantine Monetary Economy*, pp. 450–8.

[126] *Late Medieval Mints: Organisation, Administration and Techniques, BAR* International Series 389 (Oxford, 1988), edited by N. J. Mayhew and P. Spufford, *passim*.

[127] See n. 107.

[128] R. Meiggs and D. Lewis, *A Selection of Greek Historical Inscriptions to the End of the Fifth Century BC* (Oxford, 1969), pp. 111–17, no. 45, clause 5.

[129] Cicero, *ad Att.* viii. 7, 3.

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Crawford explains away the possibility of drawing money from the mint by suggesting that Cicero had money on deposit there, but the only evidence for private deposits being held at the mint is based on a modern restoration of a corrupt passage in another letter of Cicero.[130] If individuals could have coinage struck by the mint in certain circumstances it would help to explain why 'aedilician' types sometimes refer to games which were funded to a significant degree by private expenditure.[131] In particular it would make easier to comprehend the (seemingly) natural link between the massive issues of the curule aediles M. Aemilius Scaurus and P. Hypsaeus in 58 BC and the lavish aedilician games of the former, which must have entailed considerable private as well as public expenditure.[132] It was on the grounds that 'neither evidence nor likelihood supports the view that in normal times an issue of coinage was paid for out of a private individual's pocket' that Crawford was forced to reject the link between the exceptional coinage and the exceptional expenditure in the same year. Mattingly was right to point out, however, that expenditure by aediles on games which it was their duty to hold can only be defined as private in a limited sense.

In the fourth century AD we are on firmer ground. We know that gold coinage had been struck for private individuals because a law of AD 369 addressed by Valentinian I, Valens and Gratian to the *Comes Sacrarum Largitionum* ordered the confiscation of such coinage.[133] This may have proved impossible to implement as in AD 374 confiscation was mitigated to an enforced contribution of two ounces in the pound (16·7 %), calculated on the basis of what had been entered in the accounts (*breves*).[134] We do not know when the practice of minting for individuals started. It was surely not entirely illicit as the sums involved were entered in the accounts; it may have been unusual as the earlier law opined that individuals judged themselves worthy of condemnation if they considered that they should bring their own gold to the fiscal mints without coercion!

A further factor in the decision to create a new coinage rather than to use existing coin, and one which is also in danger of being underestimated, is pride. At Sestos when the demos decided to use its own bronze coinage the first reason given for the decision was so that the city's coin type should be current.[135] A Ptolemaic official, probably the master of the mint in Alexandria, expressed concern that he was being prevented from striking coins for individuals thus: 'For I believe that it is advantageous that as much gold as possible should be imported from abroad and that the royal coinage should always be fine and new, at no expense to the king.'[136] A competitive

[130] Cicero, *ad Att.* xv. 15, 1, where † me item † is restored as Monetam by D. R. Shackleton-Bailey, *Cicero's Letters to Atticus*, vol. 6 (Cambridge, 1967), p. 262.
[131] See n. 57.
[132] See n. 59; link rejected by Crawford, *RRC*, p. 707.
[133] *C.Th.* ix, 21, 7.
[134] *C.Th.* ix, 21, 8.
[135] See n. 120.
[136] See n. 107.

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desire for self-advertisement by both cities and individuals is a fundamental factor in the typology of the civic coinages of the Roman East and it is hard to believe that it was not at times also a motivation for the coinage.[137] A familiar example from literature is the petition of Alexander of Abonuteichos to the emperor to allow his city to strike coins bearing the types of his prophetic serpent Glykon and of Alexander himself, with suitable attributes.[138]

Pride shaded into politics. The despatch to Illyricum by Procopius of representatives furnished with gold coins stamped with the image of the new emperor (aureos scilicet nummos, efficiatos in vultum novi principis) may have served not only as an enticement to support him in his revolt against Valens but also as an affirmation of the status of the new emperor.[139] A number of passages in the *Historia Augusta* illustrate the use of coinage to demonstrate imperial status.[140] Valerian, on learning that Odenathus had recovered the east from the Persians, gave Odenathus a share in imperial power, conferred on him the name of Augustus, and ordered coins to be struck in his name. When the senators tried to acclaim Severus Alexander with honorific names they cried out 'Let the name of Antoninus be restored to the coinage' (Monetae nomen Antonini reddatur). There is no need to enter the debate about the standing of the *Historia Augusta* as history in order to accept that such passages could be construed in the thought world of the time of its composition, that is at the end of the fourth century. Affirmation of status by coinage can be seen inverted in the withdrawal of coinage in the name of damned or unrecognised emperors. This practice is demonstrated for the principate by the senate's withdrawal of the bronze of Gaius and by Caracalla's melting down of coins of Geta, both recorded by Dio.[141] Licinius is said to have melted the coins which Constantine had struck in honour of his Sarmatian victory.[142] In the fifth century John Cassian states that it was one of the tasks of moneychangers to recognise the coinage of tyrants.[143]

Important as some of the motivations to coin already discussed may have been, the real taboo, according to Crawford's thesis, is to suggest that

[137] Howgego, *GIC*, p. 91; K. Harl, *Civic Coins and Civic Politics in the Roman East A.D. 180–275* (Berkeley, 1987), p. 32, on coin types which reflected the claim to fame of individuals.

[138] Lucian, *Alex.* 58; L. Robert, *A travers l'Asie Mineure* (1980), pp. 410–11.

[139] Ammianus 26, 7, 11.

[140] *Gallieni Duo* 12, 1; *Severus Alexander* 8, 3; cf. *Diadumenian* 2, 6; *Tyranni Triginta* 26, 2; 31, 3.

[141] Dio lx, 22, 3; cf. Statius, *Silvae* iv, 9, 20–3 (*asse Gaiano*). For withdrawal of the gold and silver of Gaius: A. M. Burnett, 'The authority to coin in the late Republic and early Empire', *NC* 137 (1977), pp. 37–63, at p. 55. Coins of Geta: Dio lxxviii, 12, 6. For further references to *damnatio memoriae* see Howgego, *GIC*, pp. 5–6, with notes 30–1.

[142] *FHG* 4, p. 199, 14.

[143] John Cassian, *Collationes* 1, 20; for an earlier expression of adverse sentiment against the coinage of a tyrant (Nero): Arrian, *Epictetus* iv, 5, 17–18.

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coinage was ever struck in order to provide a means of exchange. But is it really so unreasonable to believe that coin was on some occasions struck to facilitate private trade or the payment of taxes, or to meet a public need for coinage? Some ancient evidence suggests that it is not unreasonable at all.

Here views on coinage expressed in literature are important and show that ancient authors thought that coins should be, and indeed were, struck to facilitate exchange.[144] Ephorus, the fourth century BC historian, said that silver was first coined in Aegina because it was a trading centre (ἐμπόριον) and because, as a consequence of the poverty of the soil, the people engaged in trade by sea.[145] Demosthenes in a speech of 353/352 BC attributed to Solon the belief that coinage was invented by individuals for transactions with individuals.[146] What is significant is not whether Ephorus and Demosthenes were right about the purpose behind the earliest production of coins but, given that they must have known why coins were struck in their own day, that these motives seemed reasonable to them.

This line of argument is supported by a number of philosophical passages. In Plato's *Republic* it is argued that a state needs a marketplace and coinage to facilitate exchange.[147] In a passage of the *Laws* concerned with constraining the acquisition of wealth in a hypothetical state it is envisaged that individuals should have a local currency for daily exchange, in order to fulfill the necessities of dealing with artisans and paying wages. This local currency should not be valid outside the city and common Hellenic coin should be obtained by the city for military expeditions and diplomatic missions abroad. Private persons returning to the city with common Hellenic currency would have to exchange it back into local coin.[148] Aristotle in the *Nicomachean Ethics* considers that coinage was introduced to allow the exchange of work of unequal value, for example between a doctor and a farmer.[149] In the *Politics* he argues that money was invented as a necessary consequence of the development of the import of needs and export of surpluses with the outside world.[150] Such ideas were not confined to classical Greece. The author of the *De Rebus Bellicis* of the fourth century AD begins his brief history of coinage with reference to the convenience of buying and selling (emendi et vendendi utilitas) and later attributes the striking of bronze to the requirements not only of military donatives but also of various civil transactions (varia populorum commercia).[151] In the sixth century Cassiodorus argued that integrity was required above all in the management

[144] The Greek evidence has been discussed in an attempt to disprove any connection between coinage and sovereignty by T. Martin, *Sovereignty and Coinage in Classical Greece* (Princeton, 1985), pp. 214–18.

[145] Cited by Strabo viii, 6, 16 (376).

[146] Demosthenes xxiv, 213 (766).

[147] Plato, *Republic* ii, 371b, 8–9.

[148] Plato, *Laws* v, 741e–742c.

[149] Aristotle, *Ethics* v, 1133a, 17–20.

[150] Aristotle, *Politics* i, 1257a, 19–41.

[151] *De Rebus Bellicis* i, 6; i, 9; in *De Rebus Bellicis*, ed. M. Hassall and R. Ireland, BAR International Series 63 (Oxford, 1979), part 2, p. 5.

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of the mint where the emperor's image was impressed and wherein was to be found a source of general utility (generalis utilitas).[152] Taken together, these passages in ancient literature do not merely illustrate the commonplace that coinage was a means of exchange but show that their authors considered that the needs of transactions within and beyond the state provided a reason for coining.

That concern to facilitate the payment of taxes could be a reason for issuing certain types of coin is nicely demonstrated by an anecdote in the life of Severus Alexander in the *Historia Augusta*.[153] Taxes which had been ten aurei under Elagabalus were reduced to a third of an aureus by Alexander, after which coins of a third of an aureus (*tremisses*) were struck. Alexander also coined quarter aurei, which he kept in the mint until he could again reduce the tax. When the needs of the state prevented him from reducing the tax as intended, he ordered that the quarter aurei be melted down. There is no possibility of this tale being historical but what is significant is that it could be written at the end of the fourth century.

Coinage was a matter of public concern and, if problems arose, protests could bring about executive action. An appeal by traders, small stall-holders and fishmongers against abuses by moneychangers at Pergamum extracted a reply from Hadrian, or a proconsul of Hadrianic date.[154] When in *c.* 85 BC the coinage at Rome was so volatile that no one knew what it was worth the praetor M. Marius Gratidianus took the initiative in proclaiming an edict on his own. The populace were so grateful that the praetor was honoured with statues at every street corner, and incense and candles were offered to him.[155] Despite the element of exaggeration the point is made.

Action could be taken, and was taken, to improve the supply of coinage in circulation. In the periodic coin shortages of the late Republic and early Principate we hear of Cicero preventing the export of gold and silver from Puteoli in 63 BC (there had been numerous earlier *senatus consulta* forbidding the export of gold), of Caesar reviving measures which limited the possession of coins by individuals to 60,000 sestertii in 49 BC (again not the first time this measure has been taken), and of Tiberius providing interest free loans in AD 33.[156] Crawford dismissed the last measure as taken for the preservation

[152] Cassiodorus, *Variae*, 7, 32, 1. [153] *Severus Alexander* 39, 6–10.

[154] A. Macro, 'Imperial provisions for Pergamum: *OGIS* 484', *GRBS* 17 (1976), pp. 169–79.

[155] Cicero, *de Off.* iii, 80; Pliny, *HN* xxxiii, 46 (132).

[156] Measures against export: Cicero, *in Vatinium* 5 (12); *pro Flacco* 28 (67). The measures in Italy were general; the application to the tax paid by Jews to the Temple in Jerusalem was an aberration of Flaccus in Asia: A. J. Marshall, 'Flaccus and the Jews of Asia (Cicero, *Pro Flacco* 28, 67–69)'; *Phoenix* 29 (1975), pp. 139–54; E. M. Smallwood, *The Jews under Roman Rule from Pompey to Diocletian* (Leiden, 1976), p. 126. The general measure of 63 BC may be connected with the shortage of coin (Valerius Maximus iv, 8, 3), the earlier enactments may have been concerned with extravagance, compare the censure of the export of money outside the empire by Tiberius (Tacitus, *Annals* iii, 53, 5) and the Elder Pliny (*HN* xii, 41, (84); cf. vi. 26, (101)). Limits to the possession of coin by individuals: Dio xli, 38, 1 ff. Loans: Tacitus, *Annals* vi, 16–17.

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of social status rather than 'for economic reasons'.[157] The inability of the élite to obtain coinage in order to settle debts was indeed one of the mechanisms which might force a government to supply coinage; it was not the only one.

Measures to control the loss of coin from circulation were also taken in the fourth century. A law of AD 374–5 from the east forbade the export of gold to the barbarians and stipulated, rather intriguingly, that any gold found among them be removed by subtle ingenuity.[158] A law of AD 354 or 356 despatched to Arles in Gaul forbade the melting down or transport of coins to different regions in order to sell them (*negotiatores* were allowed to carry a thousand folles on their animals for expenses).[159] This suggests a concern for the availability of coin in certain areas of the empire, which might have arisen, for example, from an inability to extract the payment of taxes in coin.

We have seen that ancient authors believed that coinage was produced in order to facilitate exchange by individuals within and between cities, and to enable taxes to be paid, that public concern about the failure of a coinage to function smoothly as a means of exchange could bring about executive action, and that measures were taken to improve the supply of coinage. Thus, despite the lack of explicit evidence that a state reacted to a specific shortage of coins for private transactions by striking new coin, the converging evidence cited above makes it reasonable to consider that thought for the supply of coinage could have been a factor in the decision by ancient states to strike coins.

CONCLUSION

The evidence for certain aspects of the monetary history of the ancient world which has been considered above renders untenable the proposition that the sole purpose for the issue of coins in the ancient world was to enable states to make payments. That it was an important purpose is not contested, although it is suggested that the importance of non-military expenditures may have been underestimated and that it is not always possible to comprehend patterns of coinage output without knowing how the necessary bullion was obtained. The link between coinage and expenditure is weakened by a consideration of periods of expenditure which did not call forth new coinage, and by the use of old or foreign coin, credit, or bullion to make payments. However, the validity of expenditure as a general explanation of coinage is called into question most of all by the demonstration of other types of motivation. These included not only technical monetary factors, such as re-coinages connected with monetary reforms or closed currency systems, the

[157] Crawford, *JRS* 60 (1970), p. 46.
[158] *C.J.* iv. 63, 2.
[159] *C.Th.* ix. 23, 1; Hendy, *Studies in the Byzantine Monetary Economy*, pp. 291–4.

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need to renew worn coin or to standardize a disparate accumulation, and, perhaps, the terms of monetary alliances, but also the substantial concerns of profit, pride and politics. It is argued that the practice of minting by the state on behalf of individuals (in which state expenditure has no explanatory power) can be demonstrated at least for certain periods of antiquity. Furthermore, the denial by Crawford that coined money had in the ancient world an economic reason for existence seems to run counter to the evidence for measures taken in reaction to popular pressure to ensure the smooth functioning of the currency as a means of exchange, and fails to explain adequately steps taken to improve the supply of coinage at times of shortage. It is the clear testimony of ancient authors that coins could be struck to facilitate exchange between individuals, the payment of taxes and external trade.

In considering the motives of ancient states for producing coin we have to return to an uncomfortable pluralism from the austere but comfortingly tidy notion of Crawford. State expenditure may indeed have been an 'important' motivation but we cannot know exactly how important it was or which classes of coinage were typically connected with state expenditure. The problem is that, once other types of motivation for the issue of coins are admitted, we have no firm criteria for answering such general questions. Our answers must be confined to particular issues for which there is specific evidence.

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